SpeedBot Challenge

Angel Holguin
Cristiano Da Silva
Konrad Yee
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L2: Vex Robotics System

1. 6 different subsystems that work together.

Structure: The structure subsystem provides the physical framework for the robot, including its chassis, frame, and components. It serves as the foundation upon which all other subsystems are mounted.

Motion: The motion subsystem is responsible for controlling the movement of the robot. It includes motors, servos, gears, and sprockets that convert electrical energy into mechanical motion, allowing the robot to move its various parts, wheels, or arms.

Power: The power subsystem supplies energy to the robot. This typically involves batteries and chargers, ensuring that the robot has a stable source of electrical power to operate its components.

Sensors: Sensors are an essential part of the robot's perception and feedback system. Sensors like bumpers, limit switches, and ultrasonic sensors detect the robot's environment and provide data to the control system, helping the robot make decisions and navigate.

Logic: The logic subsystem is responsible for processing the data from sensors and making decisions. It consists of a microcontroller that runs a program or logic to control the robot's behaviour based on sensor input and user-defined instructions.

Control: The control subsystem enables human interaction with the robot. It includes components like a joystick, receiver, and tether cables, allowing an operator to provide real-time commands and control the robot's actions. The six subsystems are structure, motion, power, sensors, logic, and control.

2. Ultrasonic sensor

An ultrasonic sensor emits high-frequency sound waves, which bounce off a reflective surface and return to the sensor. The sensor measures how long it takes for those waves to reflect off objects and return. Measuring the time delay, the sensor can determine the distance to the object, enabling the robot to sense its surroundings. This process follows the principles of echolocation, which is also how bats navigate through caves in the dark.

3. How does the installation of a sensor improve the functioning of the robot?

Installing a vision sensor improves the robot's functioning by providing visual data about its environment. This enables the robot to avoid obstacles, navigate more accurately, and control gestures. The vision sensor can also be used for orientation, targeting, and measuring the robot’s angle.

4. Communication with the robot brain

The wireless V5 controller communicates with the robot brain connected via Bluetooth to transmit instructions and receive data, allowing for real-time control and feedback. The controller communicates with the robot brain wirelessly, typically using methods like radio frequency (RF) or Bluetooth for remote control and operation.

L3: SpeedBot Build

1. **Speedbot Real World Applicability**

The Speed bot could be in various contexts in the real world. It can be used to transport items, such as boxes, throughout a warehouse. In a scientific context, it can be used to transport highly reactive or explosive material. For example, the vex bot can be used to transport an explosive to a site. An additional arm can drop off the item, then exit. The speedboat is an extremely rudimentary version of the rovers used on Mars for soil tests. Lastly, the speedboat has an educational application for teaching robotics and engineering.

2. **Applications of a Larger Bot**

If the Speedbot were 5 times larger, it would have a larger carrying capacity and require more powerful components. It would be able to carry heavier payloads and perform activities requiring more strength. However, it might become less agile and more challenging to navigate in confined spaces, it would become less agile and more challenging to navigate in confined spaces.

3. **Applications of a Smaller Bot**

If the Speedbot were 5 times smaller, it would become more compact and agile. It could navigate tight spaces with ease and be able to handle precise movements in smaller environments. However, its carrying capacity and power would be significantly reduced. It would also make the bot more susceptible to damage in small collisions.

L5: SpeedBot Challenge

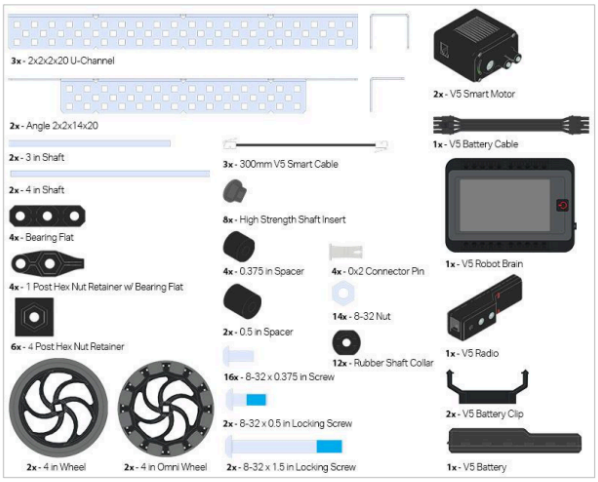
Our course was scaled 1 cm:1 ft. We did not need to combine the tracks, as ours was ultimately chosen for the final track. The scales and the units need to be the same to ensure that the tracks connect and the car has enough space between tracks to drive over. The tracks connected also need to be large enough to cover the entire 12 by 32 square feet area. It would alter how much space the car had on either side, changing the difficulty of driving along the track. If there were more space, it would be easier, and vice versa.

1. **Identify Needs**

The objective was to create a vex robot to complete a race track at a competitive speed, and then add engineering modifications that improve its performance. The vex robot car needed to have Four Wheel Drive for better acceleration, controllable steering, a decent top speed, and wheels that have proper traction. The size, materials, dimensions, and other attributes of the robot must match the guide used to create the VEX robots. The wheels should have good traction to prevent the car from drifting when steering.

The following materials and their quantities were used:

- VEX EDR V5 Classroom Starter Kit

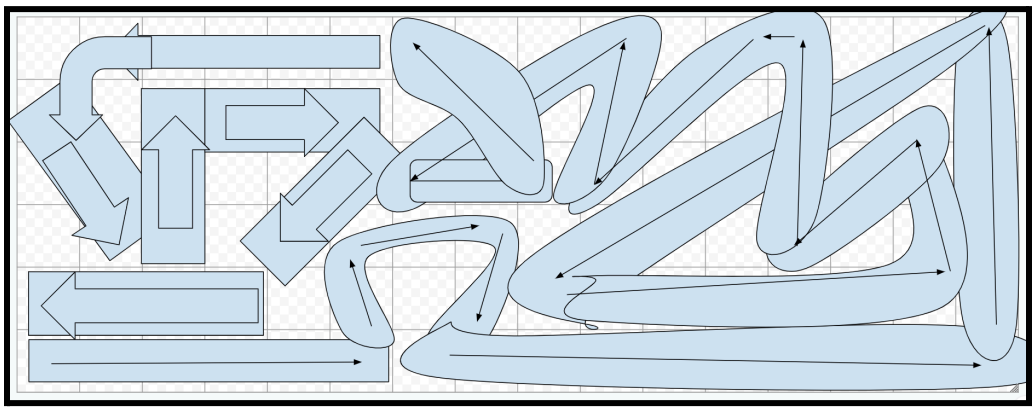


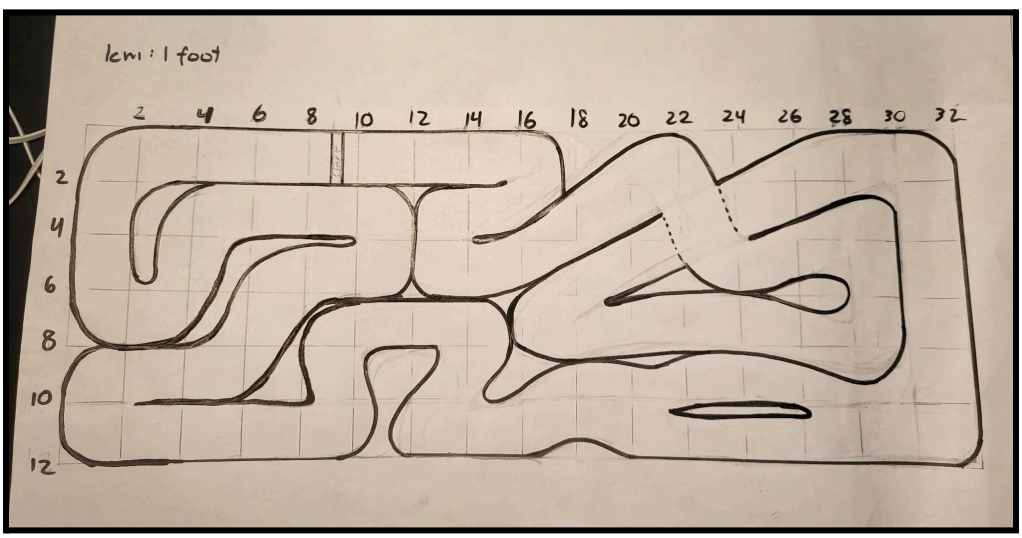
2. **Research**

Instructions for how to create the base of the robot can be found in the VEX instructions manual. For research, our group researched various wheel combinations to determine which created the optimal combination of traction and directional steering. Our group also considered the effectiveness of having Four Wheel Drive to improve the car's acceleration and speed.

3. **Generate Ideas**

We determined the following conclusions during our planning stage. First, the omnidirectional grey wheels and the smaller indented green wheels were ultimately settled upon. Second, the Four Wheel Drive would be significantly faster than a two-wheel drive. Lastly, for the race track, we generated different ideas about features we wanted to include. For example, we wanted the track to overlap itself at some point, and we featured that in the second half of the track.

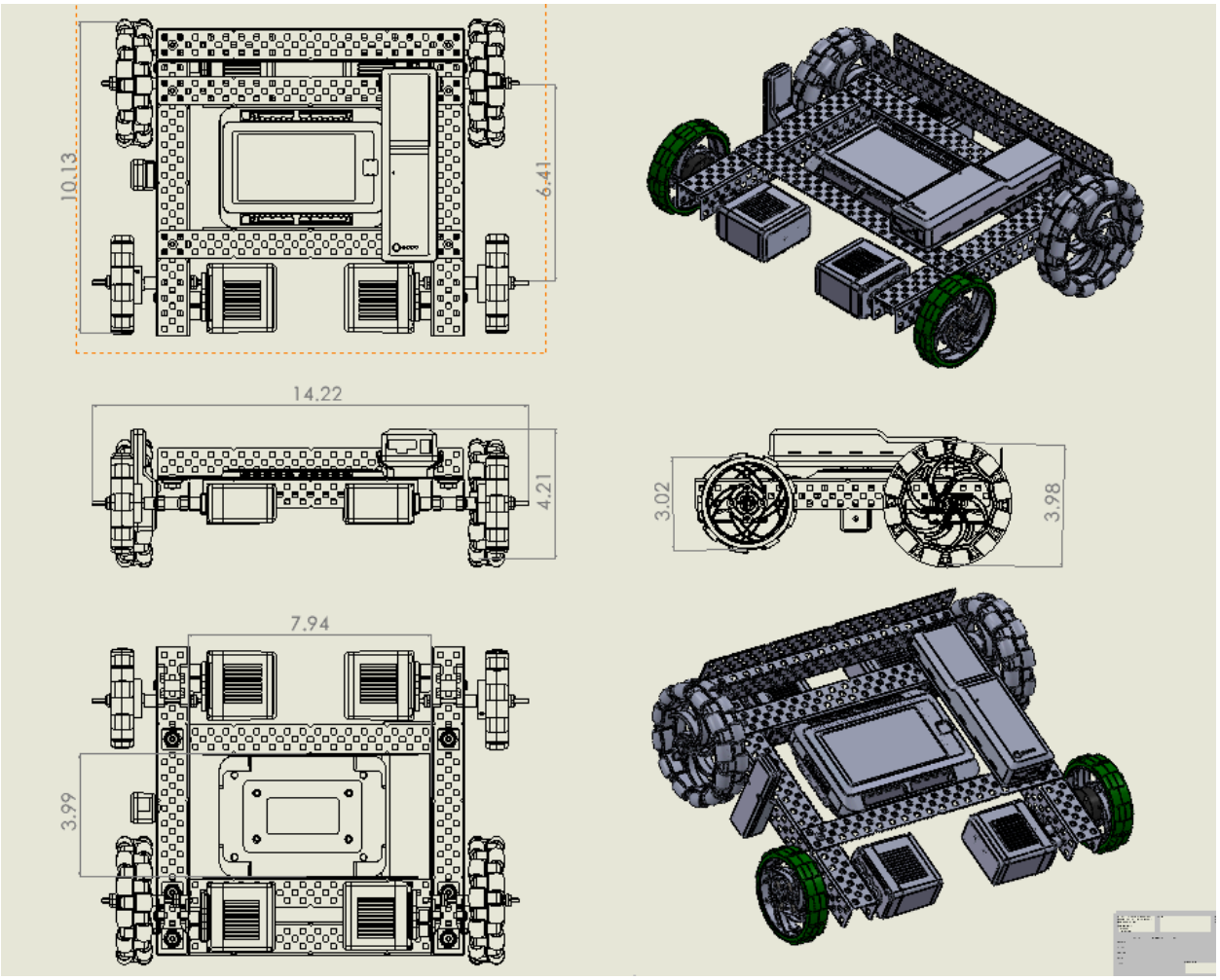




4. **Evaluate the Viability**

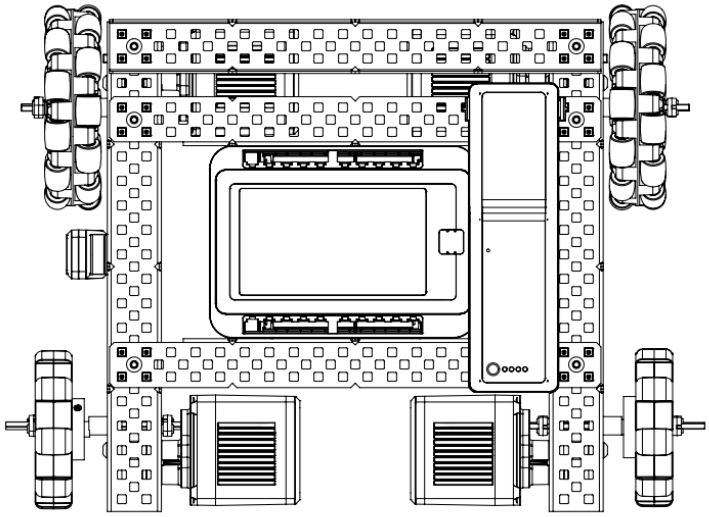
We assessed the viability of our car design by assembling our model and assessing whether the parts. For the model, our design was modified, given an original SolidWorks file by Michael Mohn on Grabcad. The link can be found in the resources section.

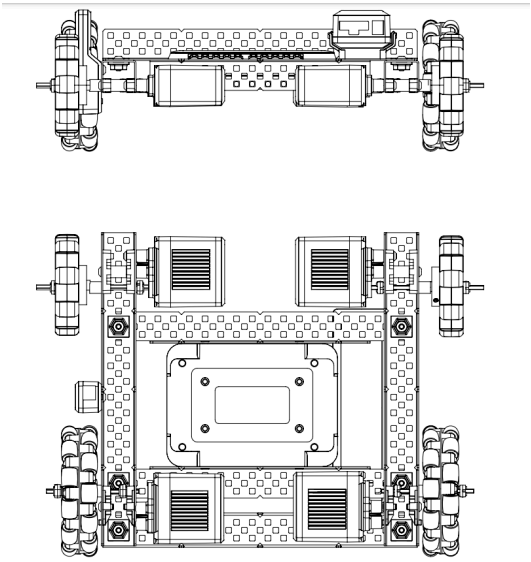
Created using SolidWorks, here is a drawing of our CAD robot after the modification. There are more designs shown in the “presentation section.”



The plan was to:

1. Find all the appropriate parts
2. Relocate the battery to the front of the car
3. Change the wheels to the omnidirectional grey wheels and the smaller indented green wheels
4. Correct for any errors in steering





5. Produce

We relocated the battery from the bottom of the car to the top, installed the two additional motors for Four Wheel Drive, and swapped the wheels. To test the prototype, we ran the car on the track several times and noticed a substantial decrease in the lap time and speed. The car veered to the left when driving straight. We attempted to correct this by changing the car's axle to one that is straighter.

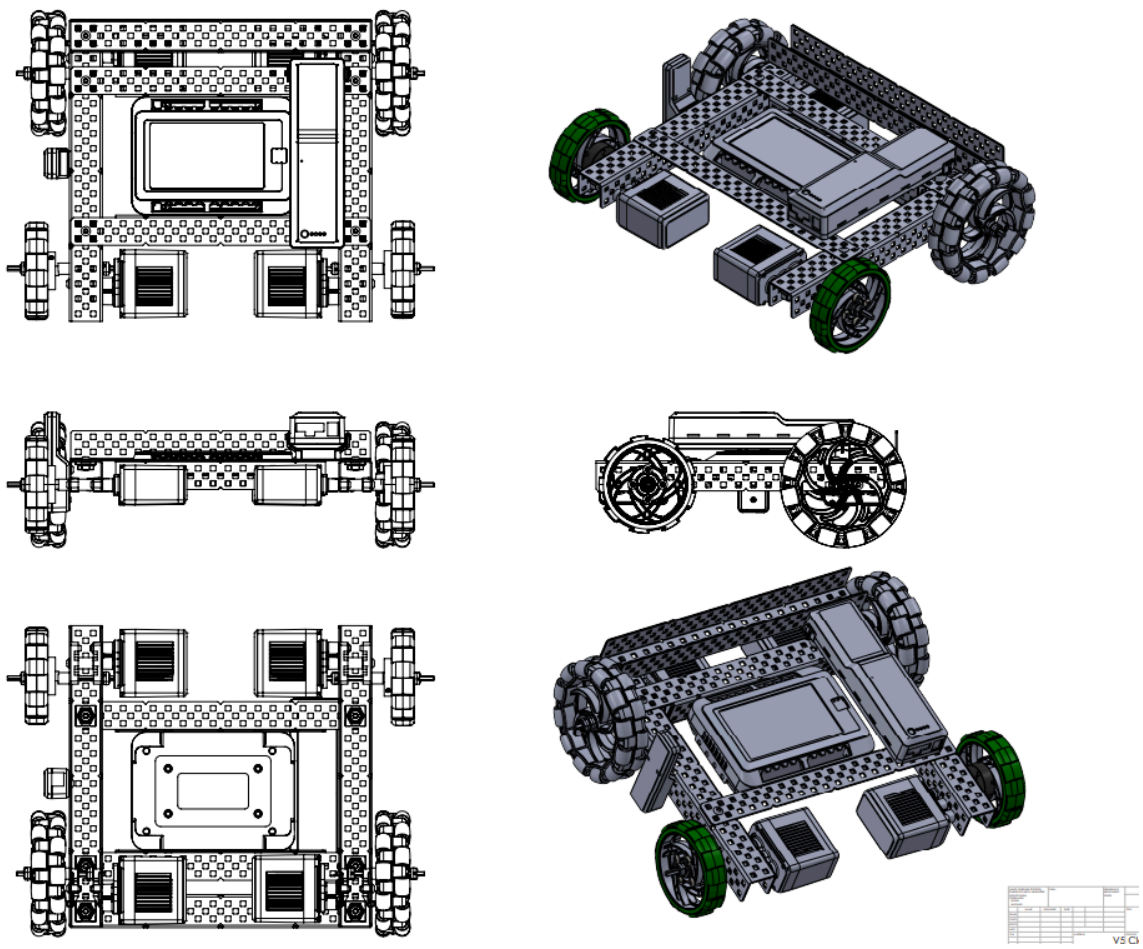
6. Test

The car was tested by competing on the race track. The car performed far better than before the engineering changes. However, driver error contributed to the added time.

Initial Time: 1:33

After changes: 1:10

7. Present



8 and 9. Present and Reflect

The car was ultimately modified and completed in less time. However, we were unable to correct for the car slightly veering to the left.

If we were to repeat this assignment, we would look further into correcting the axle to make the car drive straight. We would also change the design of the car to have a smaller frame, making it easier to navigate turns. This experiment has taught us that oftentimes it can be difficult to correct for poor car steering.

For someone about to start the same project, I would advise them to think carefully about how to make controlled steering and a car that naturally drives straight. Also, choose a driver who won't get anxious under pressure if your evaluation is based on performance.

Resources

Vex model designed by Michael Mohn

VEX Robotics V5 Clawbot | 3D CAD Model Library | GrabCAD